# Evaluation of Empirical Results

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| **Refine the Strategy** | **Does the strategy's return, risk, capacity, and trading frequency align with expectations?** Yes, broadly. We rated expected return at level 3, and the backtest produced +20.4% ROI on flat bets across 128 matches, which is moderate and consistent with that estimate. Expected risk was level 4, and the 6.0% max drawdown is actually lower than anticipated, likely because 128 matches is not enough to hit a truly bad streak. On a larger sample (full league season), drawdowns of 15-25% are realistic. Capacity (level 2) is confirmed: sports betting liquidity is limited. Frequency (level 3) is validated for league-level deployment but the World Cup alone is too infrequent.<br><br>**What caused the drawdowns, and are there methods to mitigate them?** Drawdowns were concentrated during the group stage, where upsets cluster. Saudi Arabia beating Argentina, Japan beating Germany and Spain, Cameroon beating Brazil all occurred within the first 10 days of the tournament. This pattern is structural: early-round matches have the widest skill gaps on paper but also the most motivated underdogs. Mitigation approaches include reducing position sizes during the group stage opening round, tightening the edge threshold (T_edge) for matches involving heavy favorites (where the public overprices the favorite and the model may follow suit), and applying a dedicated "upset adjustment" that scales down confidence when the model's reasoning relies heavily on aggregate goal counts without acknowledging defensive matchup dynamics.<br><br>**Would earlier or later entry/exit timing improve performance?** In sports betting, entry timing translates to when you place the bet relative to kickoff. Placing bets 48-72 hours before the match locks in odds before sharp money moves the line, potentially capturing better prices. Placing bets closer to kickoff (1-2 hours before) allows incorporation of late team news (injuries, lineup changes) but faces tighter odds as the market becomes more efficient. The optimal approach is a two-stage process: generate the model's prediction 48 hours out, then confirm the reasoning still holds with final lineup data 2 hours before kickoff, placing the bet at whichever point offers the best odds. Exit timing is not applicable in the traditional sense since bets resolve at the final whistle. On betting exchanges, in-play cash-out is possible but would require a separate real-time model, which is outside the current scope. |

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| | **Can risk be reduced or trading frequency increased while maintaining target returns?** Yes on both counts. Risk reduction: the filtered scenario (betting only on top 50% confidence predictions) produced 36.6% ROI on 64 bets with the same 6% drawdown, meaning higher selectivity directly improves risk-adjusted returns. Increasing the edge threshold from 10% to 15% would further concentrate bets on the highest-conviction predictions. Frequency increase: expanding the model to domestic leagues (Premier League: 380 matches/season, La Liga: 380, Bundesliga: 306, Serie A: 380, Ligue 1: 306, Champions League: 125) adds 1,800+ matches per year. The data pipeline and prompt format are already designed to support this. The model may need additional fine-tuning on league data to adapt to the different dynamics (home advantage is stronger in leagues than in neutral-venue World Cup matches), but the infrastructure is ready.<br><br>**Can the strategy be extended to other instruments or markets to expand opportunities?** Three natural extensions. (1) Asian handicap markets: the model's scoreline predictions can directly inform handicap bets, where the spread adjusts for team quality differences. This opens a separate market with different odds and potentially better value on closely-matched games. (2) Over/under (total goals) markets: the model predicts individual scores, so the implied total can be compared against the bookmaker's over/under line. (3) Other sports with similar structure: the prompt-based approach could be adapted to basketball (NBA), cricket, or tennis, where player-level statistics are rich and publicly available. The core architecture (structured stats in, prediction + reasoning out) transfers directly; only the data pipeline and prompt template need redesigning. |
| **Applicable Market Conditions** | The strategy performs best under the following conditions:<br><br>**High-variance environments with public sentiment bias:** World Cup group stage matches, early rounds of continental tournaments (Euros, Copa America), and league matches involving popular "big name" clubs where casual bettors inflate the favorite's odds. These are the conditions where narrative bias is strongest and the gap between public perception and statistical reality is widest.<br><br>**Matches with significant squad composition changes:** When a team has recently integrated new players from domestic leagues (common between tournament cycles), historical team-level records become less predictive, but current player-level stats remain informative. The model's player-level approach has an advantage here because it adapts to the current roster rather than relying on stale team-level priors. |

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| | **Conditions where the strategy underperforms:** Tightly-matched knockout stage matches decided by margins outside the model's resolution (penalty shootouts, individual moments of brilliance, red cards). Also matches in leagues with limited player data coverage, where the input statistics are sparse or unreliable. |
| **Risk/Position Management** | **Position control:** Fractional Kelly criterion (f = 0.25) for position sizing, hard cap of 3% of bankroll per individual match, and 15% maximum total exposure across all unresolved bets.<br><br>**Stoploss/stopgain:** Drawdown circuit breaker at 20% of bankroll halts all activity for review. Consecutive loss limit of 8 pauses betting regardless of drawdown level. No stopgain is set because the strategy is designed for sustained long-term operation, but a seasonal P&L review determines whether to continue, recalibrate, or pause.<br><br>**Hedging:** Limited applicability in single-match betting. On betting exchanges (Betfair), a position can be partially hedged by laying the opposite outcome if pre-match information changes after the bet is placed (e.g., key player injury announced after bet placement). This is the only scenario where hedging is used.<br><br>**Risk red flags:** Liquidity risk: smaller leagues and lower-profile matches have thin markets where large bets move the line. Systemic risk: a corrupt match or widespread match-fixing in a league invalidates the statistical basis of the model. Model drift: as betting markets become more efficient (more quantitative bettors enter), the edge from narrative bias may shrink over time, requiring periodic model retraining and threshold recalibration. Reasoning audit failures: if the model begins producing factually incorrect reasoning chains (citing players not on the squad, referencing wrong statistics), this signals model degradation and triggers a retraining cycle. |
| **Execution Risk Assessment** | **Order fill rate:** Sports betting has near-100% fill rates at posted odds, unlike financial markets. When you place a bet at the listed odds on a sportsbook, it is accepted at that price. On betting exchanges, fill is not guaranteed for large amounts, but for our target bet sizes ($500-$2,500), liquidity on major matches is sufficient.<br><br>**Execution latency:** Not a meaningful factor. Bets are placed hours before kickoff, not milliseconds before a price move. The relevant latency is between the model generating a prediction and the human completing the reasoning audit, which should be under 30 minutes. |

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| | **Slippage / odds drift:** This is the primary execution risk. Odds can shift between prediction time (48 hours pre-match) and bet placement. Sharp money from professional bettors often moves lines closer to true value as kickoff approaches. Mitigation: compare odds at placement time against the odds at prediction time. If the line has moved more than 5% against us (reducing our expected edge below T_edge), skip the bet. Paper trading will quantify typical odds drift magnitude.<br><br>**Market impact:** At target bet sizes ($500-$2,500), market impact on major sportsbooks is negligible for World Cup and top league matches. Impact becomes relevant above $10,000-$20,000 per bet, which is well above the strategy's position limits.<br><br>**Leverage:** No leverage is used. All bets are funded from the bankroll. The maximum loss on any bet is the stake, which is capped at 3% of bankroll.<br><br>**Tax:** Sports betting winnings are taxable in most jurisdictions. In the US, net gambling winnings are taxed as ordinary income. The effective ROI after tax depends on the individual's tax bracket. At a 24% federal rate, the post-tax ROI on the full-coverage backtest drops from +20.4% to roughly +15.5%. This should be factored into the Kelly calculation as a drag on expected returns. In some jurisdictions (e.g., UK), betting winnings are tax-free for the individual bettor. |

**Table 4: Evaluation of Empirical Results**